

SPE-173508-MS

Safety Culture in Oil and Gas Operations: A Risk Analysis Framework to Address Communication and Interoperation of Multiple Interacting Organizations

Maryam Tabibzadeh and Najmedin Meshkati, University of Southern California

Copyright 2015, Society of Petroleum Engineers

This paper was prepared for presentation at the SPE E&P Health, Safety, Security, & Environmental Conference – Americas held in Denver, Colorado, USA, 16–18 March 2015.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

According to a most recent (June 2014) report by the US Chemical Safety Board (CSB) that investigated the BP Deepwater Horizon (DWH) blowout, inadequate management systems and poor safety culture were major underlying causes of that blowout. This analysis has been corroborated by many other seminal reports on the DWH accident. In fact, lack of safety culture has been identified as a common contributing cause of many accidents in the oil and gas industry.

In response to the above serious fact, there are strong recommendations that the Bureau of Safety and Environmental Enforcement (BSEE) of the U.S. Department of the Interior, as the responsible regulatory agency for offshore operations in the Outer Continental Shelf, has to promote safety culture. In addition, the BSEE director, in his speech for the fourth anniversary of the DWH accident in April 2014, emphasized the importance of fostering a culture of safety among all involved in offshore operations so that it becomes part of the way business is conducted.

Safety culture can be defined through its five primary composing elements/pillars, characteristics or sub-cultures, which include an informed culture, a reporting culture, a learning culture, a just culture, and a flexible culture. Reporting culture, as one of the five pillars of safety culture, creates organizations in which people are prepared to report their errors and near-misses. In this regard, interoperation and interaction of multiple organizations play a critical role.

This paper, by focusing on the foregoing reporting pillar, introduces a risk assessment framework to analyze multi-organizational interactions with the focus on ineffective communication. This framework models the interactions of involved organizations and key players in four different layers: the level of meta-system interactions, bi-lateral organizational interactions level, bi-lateral work interactions level, and the level of work site operations interactions.

Our extensive literature review indicates that few studies have applied risk analysis to interactions among multiple organizations, with even more limited application to the oil and gas industry, particularly offshore operations. We further show the viability of our framework by using it to develop a risk assessment model based on the investigation of the DWH accident and to analyze the interactions of key players in that case.

1. Introduction

The world is filled with uncertainties and risks, which can be expressed as product of two elements: the likelihood of failure and associated consequence(s) (Bea, 2011). This risk incorporates three main categories of accidents, illustrated in Fig.1, dependent on the frequency of occurrence and the magnitude of loss (Rasmussen, 1997).

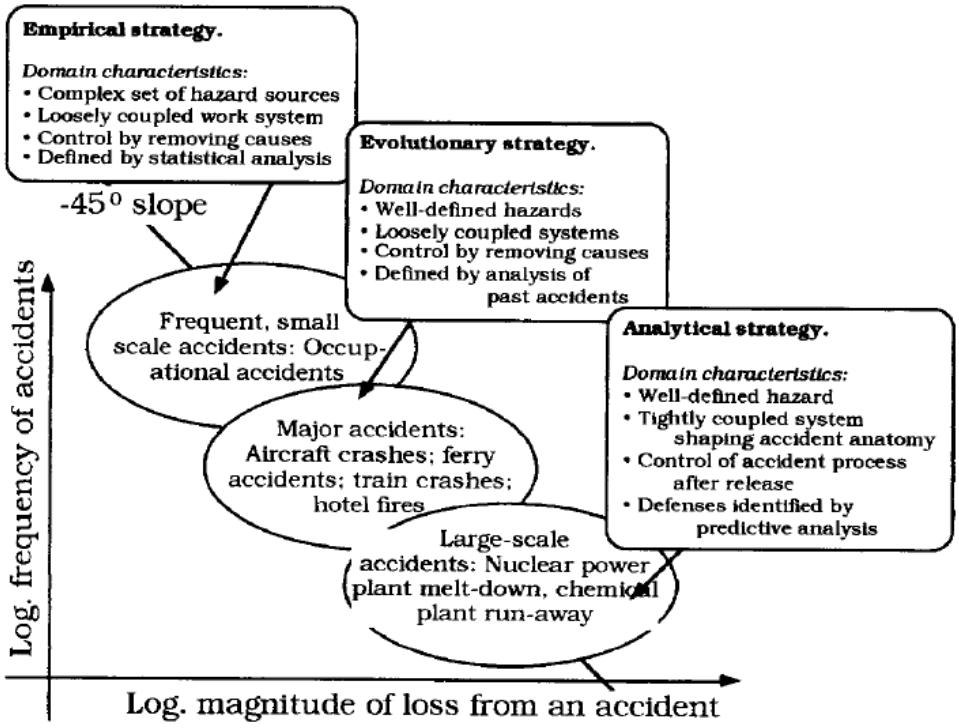


Fig. 1. Hazard source characteristics and risk management strategies (Rasmussen, 1997)

This paper focuses on so called "high-risk" systems, where large-scale accidents of low likelihood of occurrence but high level of severity or magnitude of loss occur. These systems are typified by several main characteristics. First, the technology they use changes very rapidly at the operative level (Rasmussen and Svedung, 2000). Second, they have complex interactions resulting from unfamiliar or unexpected sequences of events that are often either imperceptible or not immediately comprehensible (Wang, 2008). Third, they are tightly coupled indicating their components are highly interdependent and tightly linked together (Perrow, 1984). An important aspect of such systems is that the interactive complexity and coupling in such systems are particularly sensitive to perturbations (Leveson, 2004 and 2011) where changes in one part of the system rapidly affect the status of the others, often with disastrous results (Wang, 2008).

Adding to the complexity of large-scale accidents are interactions among multiple systems or organizations involved in those accidents where lack of coordination and communication between controllers and decision makers are important sources of unsafe behavior (Leveson, 2011).

The oil and gas industry, and the offshore operations as a specific case, is an example of a tightly coupled and interactively complex system in which different organizations interact with each other to conduct symbiotic operations. As a result, accidents in this industry can occur when key players fail to interact, as part of management failure and compromise on safety culture (e.g. Hopkins, 2008 and Goh et al., 2012).

In response to the above serious fact, different credible references recommended that the Bureau of Safety and Environmental Enforcement (BSEE) of the U.S. Department of the Interior, as the responsible regulatory agency for offshore operations in

the Outer Continental Shelf, has to promote safety culture. According to the Transportation Research Board Report (2012, page 69), BSEE has to "encourage a culture of safety so that individuals know the safety aspects of their actions and are motivated to think about safety".

It is noteworthy that the BSEE itself developed a safety culture policy statement (2013) in which announces its expectations from all individuals and organizations performing or overseeing activities regulated by BSEE. According to this statement, all those individuals and organizations have to establish and maintain a positive safety culture, foster in personnel an appreciation for the importance of safety and environmental stewardship, and emphasize the need for integrating safety into performance objectives in order to achieve optimal protection and production. In addition, the BSEE director, in his speech for the fourth anniversary of the DWH accident (April 2014), emphasized the importance of fostering a culture of safety among all involved in offshore operations so that it becomes part of the way business is conducted.

In general, safety culture can be defined through its five primary composing elements/pillars, characteristics or sub-cultures, which include an informed culture, a reporting culture, a learning culture, a just culture, and a flexible culture (Reason 1997, 1998, and 2000). Reporting culture, as one of the five pillars of safety culture, creates organizations in which people are prepared to report their errors and near-misses. In this regard, interoperation and interaction of multiple organizations play a critical role.

In another seminal reference, the United Kingdom Health and Safety Executive (HSE) defines the safety culture of an organization as "the product of individual and group values, attitudes, and perceptions, competencies, and patterns of behavior that determine the commitment to, and the style and proficiency of, an organization's health and safety management" (Presidential Commission Report, 2011, page 218).

The term "safety culture" was first applied to nuclear power by the International Atomic Energy Agency (IAEA) to explain how the lack of knowledge of risk and safety by operators and the organization contributed to the Chernobyl disaster (IAEA safety report, 1986). Although the term of safety culture is open to interpretation (Guldenmund, 2000), the concept is focused on the idea that where safety culture is practiced, and whether individuals and organizations give emphasis to safety over competing goals such as production or costs.

The BP Deepwater Horizon blowout, of April 2010 in the Gulf of Mexico, is a case study of large-scale accidents in the oil and gas industry caused by the failure of industry management; in particular by a lack of safety culture and ineffective communication between BP and its contractors (Chief Counsel's Report, 2011; Christou and Konstantinidou, 2012; Coast Guard Report, 2011; CSB Report Overview, 2014; Hopkins, 2012; NAE/NRC Report, 2011; and Presidential Commission Report, 2011). This ineffective communication was a main reason for the lack of available information to decision makers in the Macondo well blowout (May, 2013), and as the Chief Counsel's Report (2011) noted, "what the investigation makes clear, above all else, is that management failures, not mechanical failings, were the ultimate source of the disaster."

Considering root-causes of large-scale accidents, strong emphasis can be placed on prevention by utilizing appropriate communicational frameworks for the analysis of interaction and interoperability among key involved organizations.

Jens Rasmussen has been one of the pioneers in considering socio-technical components in developing and analyzing accident investigation and risk management frameworks in today's dynamic society. According to Le Coze (2014), "without a doubt, the greatest influence that Rasmussen's work has had on safety science over the past 30 years lie in its ability to

produce imaginative models based on appealing illustrations and a synthetic of different concepts from multiple disciplines in relation to empirical data".

In his foundational studies, Rasmussen modeled multiple layers of socio-technical interactions within organizational hierarchies. Fig.2, as one of the main models proposed by Rasmussen (1997), illustrates a company decomposed into interactive layers. Management, staff, and work occupy the three lowest levels. Companies additionally interact with higher level regulators and associations. Government occupies the topmost level of the interactive hierarchy.

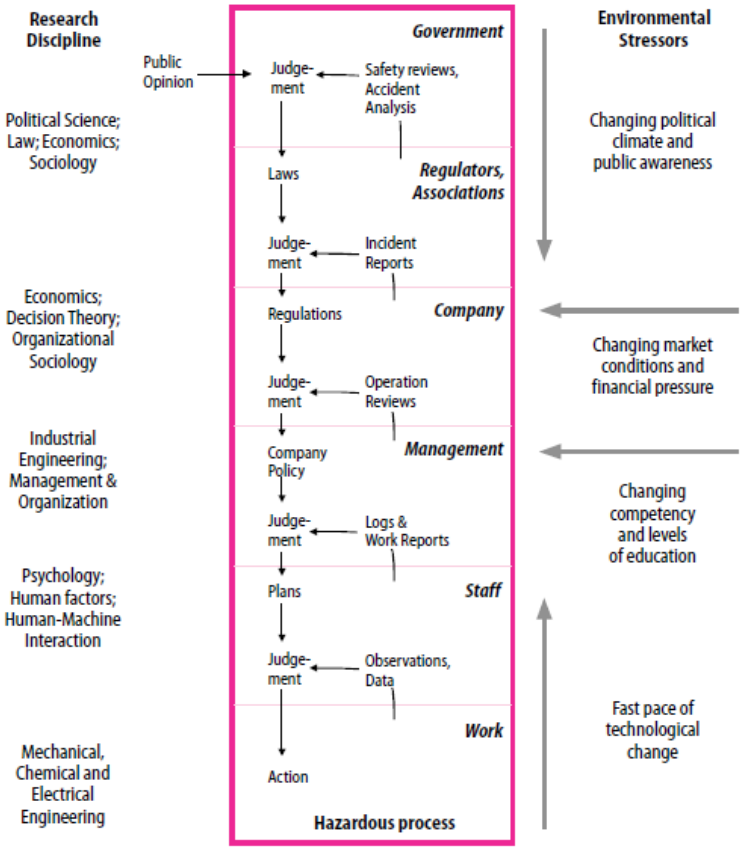


Fig. 2. Nested levels of decision making involved in risk management (Rasmussen and Svedung, 2000)

There have been other studies; some of them have been published recently; e.g. Le Coze (2014) and Trotter, Salmon, and Lenne (2013 and 2014), which used and expanded upon the above stated model by Rasmussen and other introduced frameworks by him, such as AcciMap (Rasmussen, 1997; Rasmussen and Svedung, 2000; and Svedung and Rasmussen, 2002), for better accident investigation and risk management.

The aforementioned studies have explained and utilized most of the proposed models by Rasmussen. However, there is another model originally proposed by Rasmussen in 1994, which deserves to be paid attention to as well. This model describes the concept of communication and interaction within different layers of an organization. This model has been explained in more detail in the next section.

It is our understanding that the described framework models the coordination of multiple teams and parties within only one organization. In this paper, we propose a multi-layer framework by adopting and utilizing the stated model of Rasmussen. We believe that our proposed framework has the potential to be used to analyze the interaction and interoperability among

multiple organizations involved in a work domain. It is noteworthy that this work domain can be defined in any high-risk system or industry.

In addition to the above explanation, our literature review indicates a gap in the number of studies analyzing the interactions of multiple organizations in high-risk systems; especially, in the context of oil and gas industry.

In the rest of this paper, section 2 introduces our proposed framework. Section 3 applies our framework to the case of the BP Deepwater Horizon accident, followed by a general discussion about the framework applications. Finally, in section 4, we offer conclusions about our work.

2. Methodology

The main objective of our proposed framework is to evaluate and analyze interactions of multiple key organizations in high-risk systems in order to identify relevant, key factors contributing to incidents and accidents in those systems. In this regard, the essence of accident investigation can be used to identify communication related error-prone areas. These identified areas can then be monitored in future interactions of studied organizations in order to prevent similar errors from re-occurring.

The theoretical foundations of this framework, which is a four-layer multi-organizational structure, is based on a model which was originally proposed by Rasmussen in his book in 1994 (Rasmussen, et al. 1994) and was further elaborated in the Rasmussen and Svedung (2000). Rasmussen's elegant model, which also, as Le Coze (2014) succinctly said about other Rasmussen's models that he has reviewed, is truly an "imaginative model based on appealing illustrations" that was originally developed to analyze the interactions and communications of work units, activities, and decision makers within an organization.

The Rasmussen's original model relies on the propagation of interactions between work domain "bottom up" requirements and "top-down" social practice and management style (Fig.3). Bottom-up propagation relates to functional constraints, which determine the structure and content of communication between work activities and decision makers. And, top-down stream influences the form of communication.

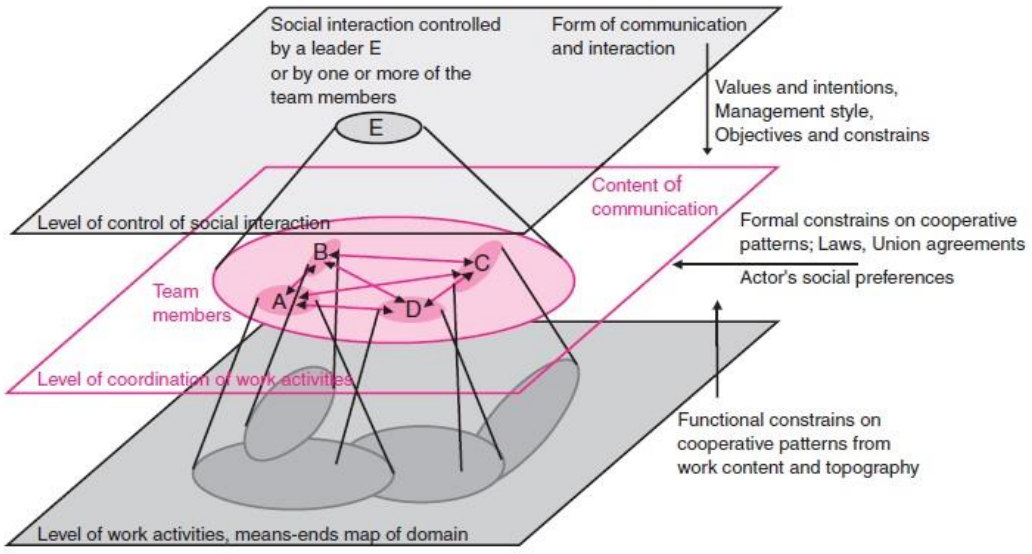


Fig. 3. A multi-layer interaction model for top-down and bottom-up coordination (Rasmussen and Svedung, 2000)

In the first layer of the Rasmussen's model, the form of communication and interaction among team members of studied organization is defined by a leader or by one or more of the team affiliates. Components such as values and intentions, management style, and objective and constraints constitute the form of communication.

The second layer of the stated model analyzes the interaction among different team members within an organization. And finally, level of work activities, which is the lowest level of work, is defined in the last layer of the model.

Our proposed framework builds on the Rasmussen's model to be able to analyze multiple organizations interactions using four main layers (Fig.4). Level I represents meta-system interactions as an integrated system, i.e. a system-of-systems, which models and manages the interaction of all the existing organizations at a social level. Level II illustrates the bi-lateral interactions of existing organizations in an identified high-risk system. Interactions in this layer are organizational.

Level III is work related and captures organizations' bi-lateral interactions at the operations level. Finally, level IV models the interactions among operators in work site operations within organizations.

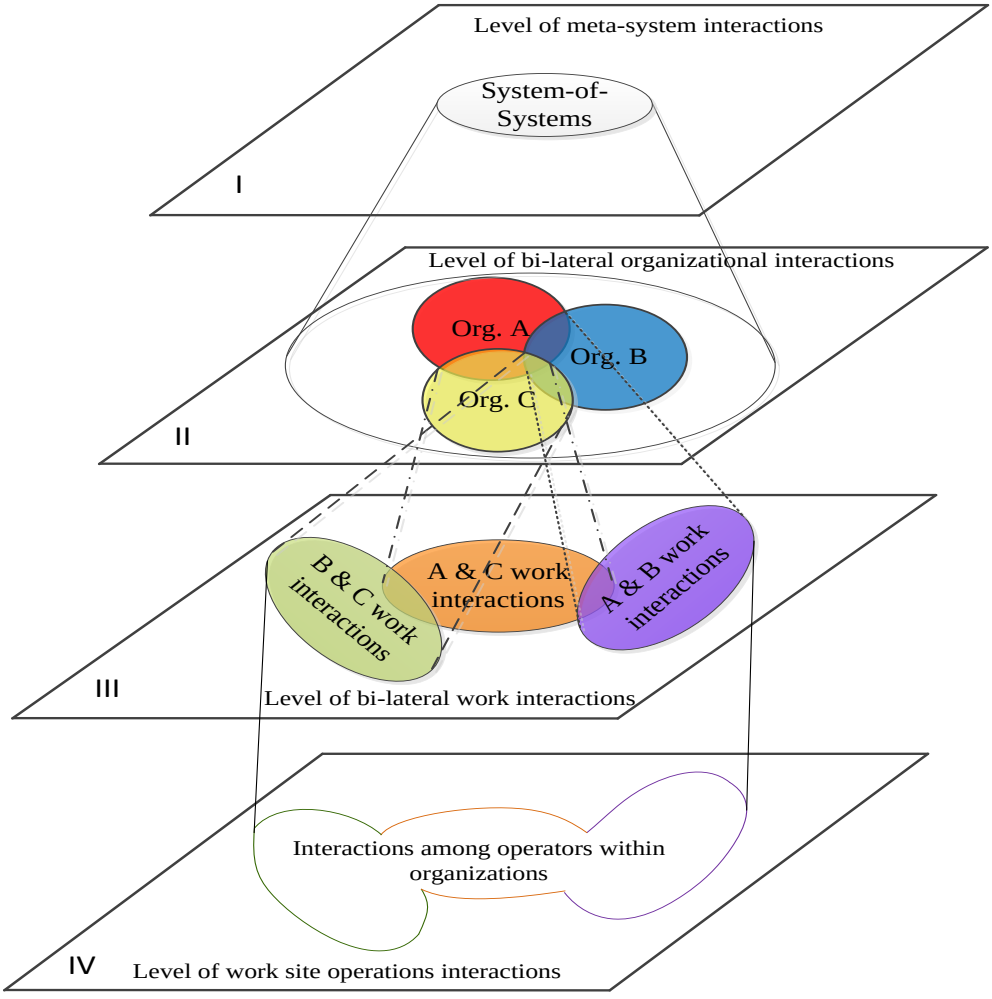


Fig. 4. Four-layer framework for risk analysis of interactions among multiple organizations

In the proposed framework in this paper, higher levels project onto lower levels. For instance, the top level projects on the second layer as a decomposition of an integrated control structure into separate bi-lateral organizational interactions. Similarly, the intersection of two organizations in the second layer is projected as a separate component onto the third level to

capture the bi-lateral work interactions of those two organizations. Finally, third layer components project onto the fourth level as interactions among operators within the identified organizations.

The first step in developing the explained framework is to identify all key players or organizations existing in a studied high-risk system. We then define the meta-system in which all key players interact. This meta-system is illustrated in the top layer of the framework. In most cases, this meta-system includes the government and related industry/ies in that context.

The second layer presents the bi-lateral organizational interactions among existing key players and captures those interactions at a high management level. These interactions are broken down into bi-lateral, work related interoperations in the third layer. The last step models interactions among lowest level operators of work activities within all the identified organizations. We will discuss the details of developing such framework and its analysis in the next section by using the BP DWH accident as a case study.

As it is illustrated and explained in this section, this framework addresses and analyzes interactions among different key involved organizations; and not the internal coordination of one organization, in different layers. In addition, it consists of the fourth layer of system-of-systems in order to include an integrated control structure of a meta-system. That is why Tabibzadeh (2014) introduced such framework and applied it to the analysis of the BP Deepwater Horizon accident.

In each layer of our proposed framework, the flow of information amongst identified key actors has to be determined and mapped. The identification of sources of information and its receivers using a communication matrix is helpful in mapping the information flow and determining the way involved organizations need to interact. In this regard, the analysis of accident scenarios for the identification of error-prone areas in different layers of communication and interaction in our developed framework is useful. This identification contributes to better monitoring different occasions of interoperation among involved actors and prevention of ineffective communication, which can lead to incidents/accidents.

Interactions of multiple organizations can be analyzed in two different situations. The first assesses interoperation of different key players in regular situations with no specific emergencies. The second analyzes communication of multiple organizations involved in emergency response situations. In either scenario, the steps from our proposed model can be used to capture the desired level of interactions among studied organizations based on the nature of an existing situation. We will elaborate using instances of ineffective communication in the case of the DWH accident for both regular and emergency response situations in the next section.

3. Case Study of the BP Deepwater Horizon Accident

As we explained in the introduction section, management failure and more specifically ineffective communication between BP and its contractors were major contributing causes of the BP Deepwater Horizon accident. The main contractors of BP in the DWH case were Transocean; as the owner of the rig, and Halliburton; as the main responsible company for cementing related services.

In this section of the paper, we intend to model and analyze interactions and communications among the three stated key players of the Deepwater Horizon case; BP, Transocean, and Halliburton, using the described framework in section 2. However, as we stated before, this is only one application of the proposed framework for the analysis of multi-organizational interactions.

Before explaining the process of developing the specified framework for modeling interactions among the aforementioned key players, we would like to highlight the critical role and importance of ineffective communication in causing the Deepwater Horizon accident. According to the 2011 Chief Counsel's Report on the DWH blowout, "inadequate communication and excessive compartmentalization of information contributed to the Macondo blowout". Some of the main stated examples of communicational issues in this report are as follows: information compartmentalization both within and between companies, not using internal technical experts effectively by not appropriately communicating risks of different operations with them, not calling shore managers to consult with them in unclear situations, and not properly sharing lessons learned from previous incidents and accidents among companies.

In addition, according to Tabibzadeh and Meshkati (2014a), issues in communication are among the first three organizational factors with the highest influence, which contributed to the Deepwater Horizon accident. Moreover, corroborating observations from other investigative studies of previous oil and gas drilling accidents indicate that ineffective communication was one of the major contributing causes of those accidents as well (Tabibzadeh and Meshkati, 2014a and 2014b).

At this stage, let us explain the process of applying the described, developed framework in section 2 for the DWH accident case study. The first step in developing this framework is to determine the components of each of the layers of the model. Based on the framework description in the previous section, the top layer comprises a meta-system or what we called a system-of-systems. The manifestation of the system-of-systems in the context of the BP DWH is the intricate and complex relation between the government and the oil and gas drilling industry. Among the oil and gas drilling industry, there are three organizations stated above that played a critical role in the Deepwater Horizon case (Fig.5).

The second layer of the framework presents the bi-lateral interactions of the three aforementioned key players in an organizational level. This level of interactions is mostly focused on communication of onshore offices of those three players with each other in order to make necessary arrangements for offshore operations.

The third layer illustrates the bi-lateral interactions of BP, Transocean, and Halliburton on the DWH rig. This level of interactions is mostly operational related. At this stage, onboard managers interact with each other in order to arrange required activities on the rig and make sure that those activities are done appropriately.

Finally, the interactions among operators of those three organizations are presented and analyzed in the bottom layer of the model. This stage of work can be expressed as the work site operations.

The next step after modeling the interactions among the stated key players in the DWH case study is the analysis of the captured interactions. The taken approach in this study is to explain the situation of captured multi-organizational interactions for the DWH case by bringing up some of the main miscommunication instances among the aforementioned players. Then, those instances are analyzed and categorized to identify the main components of interaction among those players in each layer of the framework. In addition, roles and responsibilities of each of those involved organizations are useful in determining the stated components. Those components can be used as reference points to error-prone areas in communication of analyzed key players. Finally, the development of appropriate recommendations for monitoring and tracking those areas contributes to prevention of similar accidents due to ineffective communication of involved organizations.

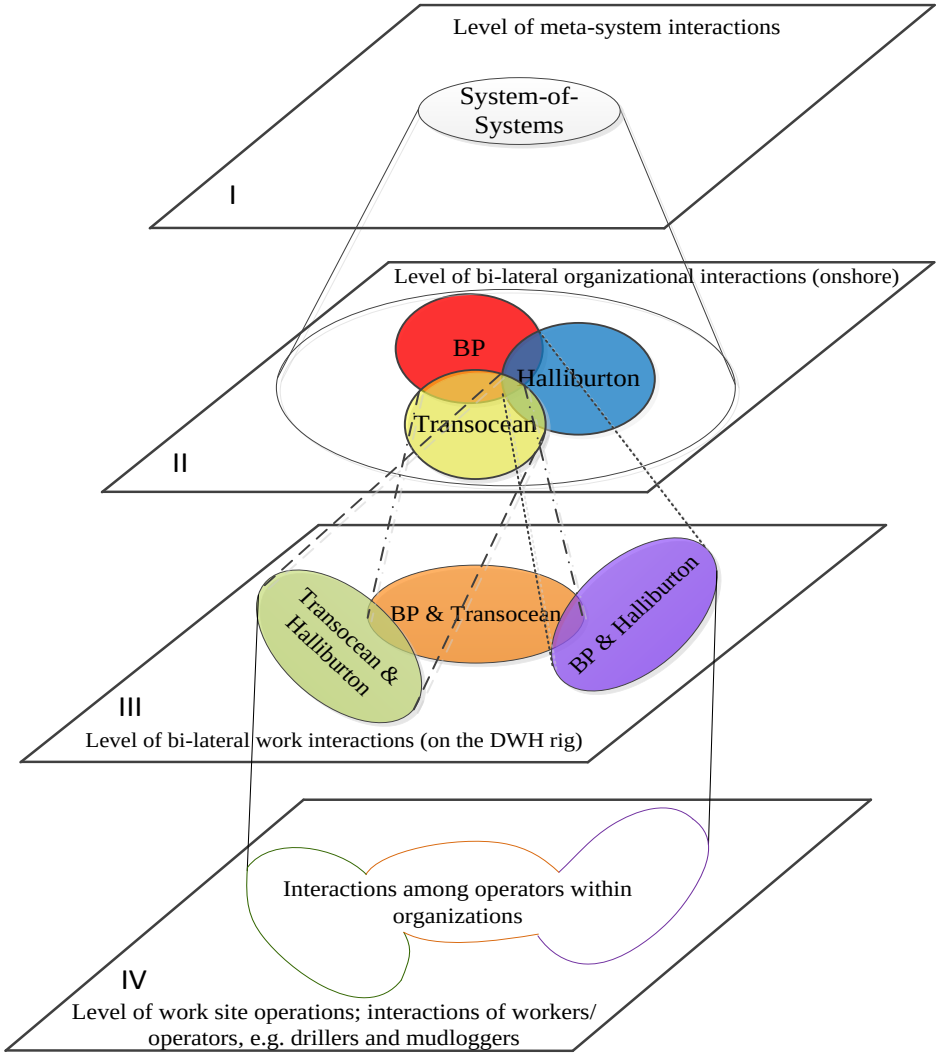


Fig. 5. Four-layer framework for risk analysis of interactions in the Deepwater Horizon case study

As stated above, the description of captured multi-organizational interactions in this section is through elaborating on some of the instances of miscommunication among the studied organizations in each of the stated layers in the proposed framework. These captured interactions in different layers of the framework have been connected to the organizational structure of the Deepwater Horizon; provided by the Chief Counsel's Report, as well in order to provide a better perspective about the interaction of BP, Transocean, and Halliburton in different instances.

As it is illustrated in Fig.6, the three main players of the DWH case communicated with each other both onshore and on the rig, which connects this organizational structure to the second and the third layer of the introduced framework in this paper. To be more specific, the bi-lateral interactions between BP and Transocean and also the communication between BP and Halliburton; both onshore, have been illustrated by dotted lines on the left section of Fig.6. Accordingly, these interactions have been modeled in the second layer of the proposed framework in Fig.5.

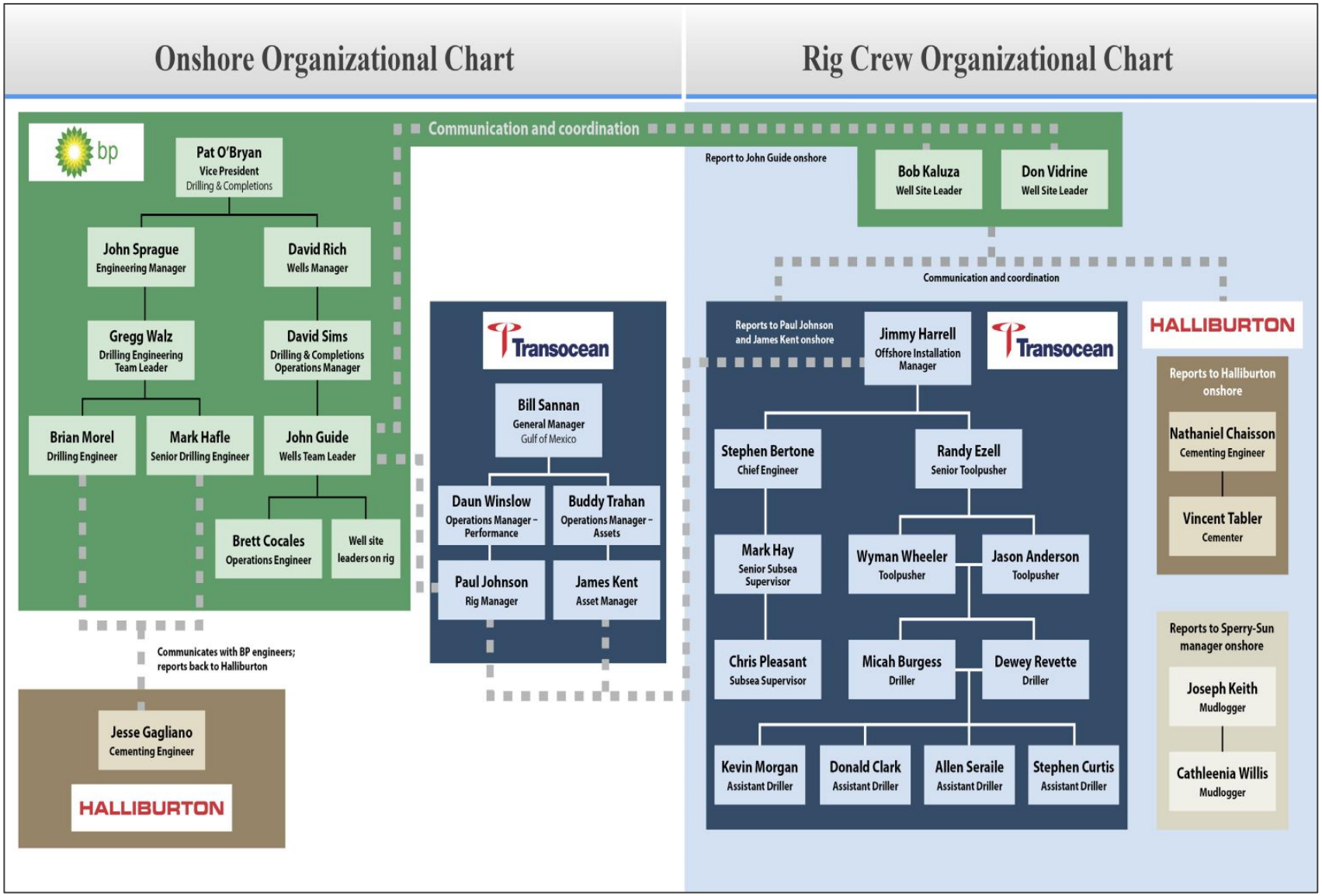


Fig. 6. Deepwater Horizon Organizational Structure (Chief Counsel's Report, 2011)

First, we analyze different instances of miscommunication between BP and Transocean in the onshore level. This will help us in indentifying the components of communication between these two players in the stated level. In one instance, BP did not effectively communicate the details of the risk management system developed by its top management with the Transocean managers. In another situation, not all lessons learned from previous similar accidents or incidents on operated rigs by Transocean were shared with BP (Hopkins, 2012). For instance, apparently, Transocean experienced a remarkably similar incident in the North Sea four months before the DWH accident while completing a well for Shell (Chief Counsel's Report, 2011 and Hopkins, 2012). Fortunately, they were able to shut in the well before any explosion by activating the Blowout Preventer (BOP). Had the Deepwater Horizon crew known about such similar incident, it was less likely for them to make decisions that led to the Macondo well blowout (Chief Counsel's Report, 2011 and Hopkins, 2012).

Other important series of miscommunication discussed in the published investigation reports for the DWH blowout were between BP and Halliburton in the onshore side. Halliburton as one of the key involved contractors in the DWH was mainly in charge of conducting cement testing and reporting relevant needed information to BP managers and engineers. One source of communication difficulties in this level of interaction is debated to be the format of Halliburton modeling reports. According to Chief Counsel's Report (2011), Halliburton could have highlighted the prediction of channeling and gas flow along with the overall assessment of cementing success in reports sent to BP engineers. This issue might have caused BP engineers to review the prediction in a cursory fashion.

In addition to onshore interactions, the crew of BP and Transocean and also BP and Halliburton collaborated with each other on the DWH rig. Those collaborations have been illustrated on the right section of Fig.6 using dotted lines. Accordingly, the indicated collaborations have been captured in the third layer of the developed framework illustrated in Fig.5.

Similar to the description of miscommunication instances in the second layer, the stated organizations had ineffective interactions in the third layer as well. One of the main discussed issues related to work operations interactions in the DWH case is insufficient communication between BP and Transocean prior to and during the final displacement, which "affected risk awareness and well monitoring" (Chief Counsel's Report, 2011). In this occasion, BP did not adequately inform Transocean about the existing risks of the Macondo well (Bureau of Ocean Energy Management, Regulations and Enforcement (BOEMRE) Report, 2011; Chief Counsel's Report, 2011; Presidential Commission Report, 2011; and Transocean Report, 2011). In addition, BP did not emphasize to the rig personnel the particular importance of the negative pressure test, as the only way to assure cement robustness in the bottom of the well (Chief Counsel's Report, 2011; Presidential Commission Report, 2011; and Transocean Report, 2011). According to Chief Counsel's Report (2011), "Had BP properly emphasized the importance of the test and the need for special scrutiny of its results, BP and Transocean personnel on the rig may have reacted more appropriately to the anomalous pressure readings and flows they observed".

As explained in the third layer of the discussed framework, BP and Halliburton collaborated with each other as well. Similarly, there were some instances of miscommunication between those two organizations. In one instance, Halliburton missed the opportunity of communicating its concern regarding overall success of the cement job in the report sent to BP (BP Report, 2010 and Chief Counsel's Report, 2011). Halliburton expressed its complete satisfaction about the cement job while later on in its post-blowout skepticism, it stated that cementing in the surface was successful whereas the success of the down-hole cementing which was uncertain.

Finally, there is a need to describe the bottom layer of the proposed framework based on the existing interactions among operators of the three stated key players on the DWH rig. One of the main instances of miscommunication analyzed in this level was between the drillers and the mudloggers. The DWH drillers were the Transocean employees and the mudloggers

were the Sperry-Sun, as a Halliburton subsidiary, personnel. Both of these groups of personnel were directly responsible for well monitoring.

Communication between the drill crew and the mudloggers broke down on several occasions. In one instance, ineffective communication between the drillers and the mudloggers contributed to the inability of the mudloggers to monitor pit levels properly (BP Report, 2010 and Chief Counsel's Report, 2011). Apparently, there were some simultaneous operations during the preparation for well suspension on the day of the accident; April 20 (BP Report, 2010). One of those operations was offloading mud to the supply vessel M/V Damon Bankston. According to different investigation reports of the DWH accident, the simultaneous operations impaired mudloggers' ability to monitor the well reliably (BP Report, 2010; Chief Counsel's Report, 2011; and Hopkins, 2011). In addition, the driller on shift did not notify the mudlogger after the offloading to the supply vessel stopped. Therefore, the mudlogger did not effectively monitor pit volumes for the remainder of the day (BP Report, 2010).

Moreover, the drill crew did not communicate with the mudlogger on shift about the anomalies such as pressure built-up or increase in the flow-out (Chief Counsel's Report, 2011). Nor did they contact any senior person available on the rig; such as the senior toolpusher, Offshore Installation Manager (OIM), or the well-site leader, in order to seek his advice.

Another critical point, which made the communication level between the drillers and the mudloggers more difficult, was the location of the mudloggers' and the drillers' shacks from each other. Those two shacks were one flight of stairs away from each other (Chief Counsel's Report, 2011). This is while both the drillers and the mudloggers were in charge of monitoring the well, which required them to be in close contact and interaction with each other.

It is noteworthy to indicate that the communication line between Transocean and Sperry-Sun on the rig has not been illustrated in the organizational structure for the DWH shown in Fig.6. This is while these two organizations needed to interact with each other for an effective well monitoring. Therefore, we suggest that the stated interaction is added to that figure, using dotted line, in order to connect the Transocean rig crew with the Sperry-Sun personnel on the rig.

As we discussed in the previous section, proper interaction of involved organizations in a high-risk system is crucial in an emergency response case as well. In the case of the Deepwater Horizon accident, there were several occasions of ineffective communication among the rig crew in the emergency response situation, which endangered life of the onboard crew. One of those instances was unclear line of communication among the well-site leader, the OIM, and the captain regarding activating the Emergency Disconnect System (EDS) after the explosion (Coast Guard Report, 2011). Because of a "clerical error" by the Republic of the Marshall Islands, the DWH was allowed to have a dual-command organizational structure in which the OIM was in charge when the vessel was latched on to the well, and the master was in charge when the Mobile Offshore Drilling Unit (MODU) was moving between locations or in an emergency situation (Coast Guard Report, 2011). However, by the start of the explosion, there was no immediate authority transfer from the OIM to the master. According to the Coast Guard Report (2011), "this command confusion at a critical point in the emergency may have impacted the decision to activate the EDS".

Ineffective coordination between the rig crew and SMIT Salvage Americas, a contractor engaged by Transocean, for firefighting efforts was another occasion of miscommunication among the involved organizations in the emergency response to the DWH explosion (Coast Guard Report, 2011).

As stated before, these instances of miscommunications need to be considered in conjunction with the roles and

responsibilities of each of the involved key players in order to understand the main types of possible ineffective interaction among those players. This contributes to identification of error-prone areas which need to be monitored in future operations for prevention of similar incidents and accidents. In the case of the DWH, knowing the main role of each of the involved key players; BP as the operator of the rig, Transocean as the owner of the rig and Halliburton as the main contractor responsible for cementing services, assists us in the described analysis process.

The analysis of the described instances of interactions indicates the critical need to provide a context for effective communication. The recommendation of the NAE/NRC Report (2011) on providing such context for effective communication on health, safety, and environmental issues among those involved organizations corroborates our finding.

As a more specific recommendation, there had to be more precise rules and regulations in defining the form, the structure, and the content of interactions among those three organizations in order to reduce the risk of miscommunication. For instance, detailed reporting as an important tool for effective communication between organizations had to be encouraged. In addition, designing standard and well-understood reporting infrastructures can potentially decrease the risk of ineffective communication. Incorporating appropriate rewarding systems for reporting bad news and unsafe situations is another policy to improve communicational issues. All these recommendations are applied for the analysis of similar future interactions in order to prevent catastrophes such as the Deepwater Horizon blowout.

Communicating the importance of critical procedures or decisions is another crucial element in effective interaction among multiple organizations. This issue results in observing and identifying risks associated with those procedures/decisions, which can be useful in preventing contributing causes of accidents.

More importantly, communication among multiple organizations has to maintain a focus on safety. Sharing knowledge and expertise across organizational boundaries is one great way of communicating lessons learned, as organizational assets. This conclusion has been corroborated by the NAE/NRC Report (2011) findings.

It is noteworthy that effective interaction among organizations is as important or even more important when involved organizations in a high-risk system deal with emergency situations. Therefore, training personnel for appropriate coordination during emergency response cases is strongly recommended. In addition, defining clear lines of communication both in regular and emergency situations contribute to prevention of ineffective interaction among involved players.

In addition to all the stated instances of miscommunication and the suggested improving recommendations, the proposed four-layer framework for the analysis of interoperations among multiple organizations can be integrated with the three-layer risk assessment framework introduced by Tabibzadeh (2014) and Tabibzadeh and Meshkati (2014b) in order to derive some preventive measures for analyzing future drilling as well as other cases of high-risk operations. The aforementioned three-layer risk assessment framework analyzes the critical role of human and organizational factors as a major contributing cause of offshore drilling accidents. Some of the main organizational factors captured in that framework are personnel management issues, issues in communication, and economic pressure.

In this section, we utilized the introduced framework in section 2 for the analysis of interactions among multiple organizations in the Deepwater Horizon accident case study. However, the DWH case study was just one example of applying the proposed framework. Almost all high-risk systems, which include tightly coupled and interactively complex operations, consist of multiple key organizations that their ineffective communication can be one of the main factors leading to large-scale accidents in those systems. Another example besides oil and gas and offshore operations is the nuclear industry.

The Three Mile Island accident, which occurred in 1979 as a result of a partial nuclear meltdown, is one of the major accidents in this industry. Inadequate communication was stated as one of the main contributing causes of that accident (Hopkins, 2001 and Walker, 2006).

Based on this argument, the proposed framework in this paper can be applied for the analysis of multi-organization interactions in any high-risk system. In another word, utilizing the introduced framework in this paper is a footstep in identifying factors associated with those interactions and in preventing occurrence of disastrous accidents.

4. Conclusion

Our study focuses on high-risk systems in which large scale accidents occur. Such systems include tightly coupled and interactively complex operations. In addition, multiple organizations operate within those systems, where their interoperation plays a critical role in the reliability and system safety of the environment in which they interact. Hence, accidents occurring in those systems can result from failures in interactions among key organizations. The Deepwater Horizon blowout is an example of a high-risk system with highly interdependent and tightly linked operations within the oil and gas industry, where management failure and more specifically ineffective communication between BP and its contractors resulted in that catastrophic event.

Analysis of this and similar high-risk systems reveals the need to develop more sophisticated frameworks within different industries to model and analyze interactions of key players in those systems, so as to prevent the factors associated with ineffective communication among these players, that lead to disastrous accidents.

In addition, our literature review indicates a gap in the number of studies analyzing the interactions of multiple organizations in high-risk systems; especially, in the context of oil and gas industry. To close this critical gap, this paper introduced a conceptual framework to analyze multi-organizational interactions and propose some recommendations for prevention of ineffective communication in future interactions of studied organizations. We then applied this framework to model and analyze interactions among BP, Transocean, and Halliburton, the three main players involved in the Deepwater Horizon blowout, to show the viability of our proposed framework and by think to contribute to the literature of safety and reliability.

References

Bea, R., 2011. Risk Assessment and Management: Challenges of the Macondo Well Blowout Disaster. Deepwater Horizon Study Group. Center for Catastrophic Risk Management, Working Paper

BOEMRE Report, 2011. Report Regarding the Causes of the April 20, 2010 Macondo Well Blowout. Bureau of Ocean Energy Management, Regulation, and Enforcement

BP Report, 2010. Deepwater Horizon Accident Investigation Report. British Petroleum

BSEE Director speech, 2014. http://www.bsee.gov/uploadedFiles/BSEE/BSEE_Newsroom/Speeches/2014/Speech-%20COS%20April%202014-%20Final%20for%20Web.pdf, Access date: December 7, 2014

BSEE safety culture policy statement, 2013. Final Safety Culture Policy Statement. 13XE1700DX EEEE600000 EX1SF0000.DSA000. Bureau of Safety and Environmental Enforcement (BSEE), U.S. Department of the Interior

CSB Report Overview, 2014. Explosion and Fire at the Macondo Well. Investigation Report Overview, U.S. Chemical Safety and Hazard Investigation Board. Report No. 2010-10-I-OS

Chief Counsel's Report, 2011. Macondo, the Gulf oil disaster. National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling

Christou, M. and Konstantinidou M., 2012. Safety of Offshore Oil and Gas Operations: Lessons from Past Accident Analysis: Ensuring EU Hydrocarbon Supply through Better Control of Major Hazards. Publication JRC77767. Joint Research Centre. European Commission,
http://publications.jrc.ec.europa.eu/repository/handle/111111111/27463, Access date: February 27, 2014

Coast Guard Report, 2011, Report of Investigation into the Circumstances Surrounding the Explosion, Fire, Sinking, and Loss of Eleven Crew Members aboard the Mobile Offshore Drilling Unit Deepwater Horizon in the Gulf of Mexico: April 20-22, 2010. Volume I. United States Coast Guard

Goh, Y.M., Love, P.E.D., Brown, H. and Spickett J., 2012. Organizational Accidents: A Systemic Model of Production versus Protection. Journal of Management Studies, 49 (1): 52:76

Guldenmund, F.W., 2000. The Nature of Safety Culture: A Review of Theory and Research. Safety Science, 34 (1-3): 215 - 257

Hopkins, A., 2001. Was Three Mile Island a 'Normal Accident'?. Journal of Contingencies and Crisis Management, 9 (2): 65-72

Hopkins, A., 2008. Failure to Learn: The BP Texas City Refinery Disaster. CCH Australia Limited

Hopkins, A., 2011. Management Walk-arounds: Lessons from the Gulf of Mexico Oil Well Blowout. Safety Science, 49 (10): 1421-1425

Hopkins, A., 2012. Disastrous Decisions: The Human and Organizational Causes of the Gulf of Mexico Blowout. CCH Australia Limited

IAEA safety report, 1986. Summary Report on the Post-accident Review Meeting on the Chernobyl Accident. International Atomic Energy Agency (IAEA), Safety Series No.75-INSAG-1

Leveson, N., 2004. A New Accident Model for Engineering Safer Systems. Safety Science, 42 (4): 237-270

Leveson, N., 2011. Engineering a Safer World: Systems Thinking Applied to Safety. The MIT Press

Le Coze, J.C., 2014. Reflecting on Jens Rasmussen's Legacy; A Strong Program for a hard Problem. Safety Science, http://dx.doi.org/10.1016/j.ssci.2014.03.015

May, S., 2013. Case Studies in Organizational Communication: Ethical Perspectives and Practices. Second edition. SAGE Publications, Inc.

NAE/NRC Report, 2011. Macondo Well Deepwater Horizon Blowout: Lessons for Improving Offshore Drilling Safety. National Academy of Engineering and National Research Council (NAE/NRC). The National Academies Press, Washington, D.C.

Perrow, C., 1984. Normal Accidents: Living with High-Risk Technologies. Basic Books, New York

Presidential Commission Report, 2011. Deep Water; the Gulf Oil Disaster and the Future of Offshore Drilling. National Commission on the BP Deepwater Oil Spill and Offshore Drilling

Rasmussen, J., 1997. Risk Management in a Dynamic Society: A modeling Problem. Safety Science, 27 (2): 183-213

Rasmussen, J., Pejtersen, A.M. and Goodstein, L.P., 1994. Cognitive Systems Engineering. First edition. Wiley-Interscience Publication. John Wiley & Sons Inc.

Rasmussen, J. and Svedung, I., 2000. Proactive Risk Management in a Dynamic Society. First edition. Raddningsverket. Risk and Environmental Department. Swedish Rescue Services Agency, Karlstad, Sweden

Reason, J., 1997. Managing the Risks of Organizational Accidents. Ashgate Publishing Limited

Reason, J., 1998. Achieving a Safe Culture: Theory and Practice. Work and Stress, 12 (3): 292-306

Reason, J., 2000. Safety Paradoxes and Safety Culture. Injury Control and Safety Promotion, 7 (1): 3-14

Svedung, I. and Rasmussen, J., 2002. Graphic Representation of Accident Scenarios: Mapping System Structure and the Causation of Accident. Safety Science, 40 (5): 397-417

Tabibzadeh, M., 2014. A Risk Analysis Methodology to Address Human and Organizational Factors in Offshore Drilling Safety: With an Emphasis on Negative Pressure Test. Doctoral Dissertation. University of Southern California, Los Angeles, CA

Tabibzadeh, M. and Meshkati, N., 2014a. A Risk Analysis Study to Systematically Address the Critical Role of Human and Organizational Factors in Negative Pressure Test for the Offshore Drilling Industry: Policy Recommendations for HSE Specialists. SPE-168559-MS. SPE International Conference on Health, Safety, and Environment, Long Beach, California, March 17–19, http://dx.doi.org/10.2118/168559-MS

Tabibzadeh, M. and Meshkati, N., 2014b. Learning from the BP Deepwater Horizon Accident: Risk Analysis of Human and Organizational Factors in Negative Pressure Test. Environment Systems and Decisions, 34 (2): 194-207, http://dx.doi.org/10.1007/s10669-014-9497-2

Transocean Report, 2011. Macondo Well Incident. Transocean Investigation Report, Volume I

Transportation Research Board Report, 2012. Evaluating the Effectiveness of Offshore Safety and Environmental Management Systems. Committee on the Effectiveness of Safety and Environmental Management Systems for Outer Continental Shelf Oil and Gas Operations. Special report 309, Transportation Research Board (TRB)

Trotter, Margaret J., Salmon, Paul M. and Lenne, Michael G., 2013. Improvisation: Theory, Measures and Known Influencing Factors. Theoretical Issues in Ergonomics Science, 14 (4): 475-498

Trotter, Margaret J., Salmon, Paul M. and Lenne, Michael G., 2014. Impromaps: Applying Rasmussen's Risk Management Framework to Improvisation Incidents. Safety Science, 64: 60-70

Walker, S. J., 2006. Three Mile Island: A Nuclear Crisis in Historical Perspective. University of California Press

Wang, H., 2008. Safety Factors and Leading Indicators in Shipping Organizations: Tanker and Container Operation. Doctor of Philosophy Dissertation. Rensselaer Polytechnic Institute, Troy, New York